COMPACT OBJECTS WITH SUPPORT ARE EXACTLY THE PERFECT COMPLEXES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Perfect complexes define compact objects of the derived category, and compactness passes to subcategories closed under direct sums. invoked at proof lines 4540-4544. Gives the easy direction: perfect implies compact in D_QCoh,T.
- There exists a generator E of D_QCoh,T(O_X) which is a perfect complex of O_X-modules. invoked at proof lines 4545-4549. Supplies the perfect generator whose compact hull controls the subcategory.
- A compact generator of a triangulated category with direct sums is a classical generator of the full subcategory of compact objects. invoked at proof lines 4550-4553. Expresses every compact object as obtained from E by shifts, finite sums, cones and direct summands.

Source reference. Original proposition 09M1: sources/stacks-perfect/pinned-original-source.tex, source lines 4464–4464. Cited at proof lines 4541-4544 for the easy direction; the transfer of compactness to the subcategory is inside the counted span.

Source reference. Original lemma 0A9A: sources/stacks-perfect/pinned-original-source.tex, source lines 4185–4185. Cited at proof lines 4546-4549 to supply the perfect generator E; the generator argument built on it is inside the counted span.

Source reference. Original proposition 09SR: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 4550-4553 to express compact objects via E; the four-operation bookkeeping preserving perfectness is inside the counted span.

Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme. Let $T \subset X$ be a closed subset such that $X \setminus T$ is quasi-compact. An object of $D_{QCoh,T}(\mathcal{O}_X)$ is compact if and only if it is perfect as an object of $D(\mathcal{O}_X)$.*

Proof. We observe that $D_{QCoh,T}(\mathcal{O}_X)$ is a triangulated category with direct sums by the remark preceding the lemma. By Proposition 09M1 the perfect objects define compact objects of $D(\mathcal{O}_X)$ hence a fortiori of any subcategory preserved under taking direct sums. For the converse we will use there exists a generator $E \in D_{QCoh,T}(\mathcal{O}_X)$ which is a perfect complex of $\mathcal{O}_X$-modules, see Lemma 0A9A. Hence

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by the above, E is compact. Then it follows from Derived Categories, Proposition 09SR that E is a classical generator of the full subcategory of compact objects of $D_{QCoh,T}(\mathcal{O}_X)$. Thus any compact object can be constructed out of E by a finite sequence of operations consisting of (a) taking shifts, (b) taking finite direct sums, (c) taking cones, and (d) taking direct summands. Each of these operations preserves the property of being perfect and the result follows. $\square$